Langston Donald Workshop Assignment

Langston Donald

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Packages

Let's start by installing/loading packages (as needed) and libraries:

```
# Load necessary libraries
library(tidycensus)
library(tidyverse)
#library(dplyr) if tidyverse is not working
#library(ggplot2) if tidyverse is not working
library(sf)
library(viridis)
library(mapview)
```

U.S. Census API key

To access the census data for mapping, please sign up for an API key using this link. You should receive an API key in your email. Place it in the code chunk as indicated below.

Introducing Your Variable

The independent variable is the presence or absence of internet access, referred to as "digital desert" or "no internet." This variable represents whether participants have reliable connectivity or experience limited to no access to the internet, showing what Fulton county and all of Georgia and what households have internet and the ones who don't.

Mapping

```
acs_vars <- load_variables(2023, dataset = "acs5")

#Total No internet access
# B28002_013E

# Total Households
# B28002_001E
```

```r
# Presence and Types of Internet Subscriptions in Household
# B28002

#100 % Households without Internet
# (B28002_013E / B28002_001E
```

```r
ga_internet_sf <- get_acs(
  geography = "county",
  variables = "B28002_001E",
  state = "GA",
  year = 2024,
  geometry = TRUE
)
```

```
## Getting data from the 2020-2024 5-year ACS

## Downloading feature geometry from the Census website.  To cache shapefiles for use in future sessions

##   |                                                                              |
```

```r
ga_internet_sf
```

```
## Simple feature collection with 159 features and 5 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: -85.60516 ymin: 30.35785 xmax: -80.84038 ymax: 35.00124
## Geodetic CRS:  NAD83
## First 10 features:
##    GEOID                       NAME   variable estimate moe
## 1  13105     Elbert County, Georgia B28002_001     8376 245
## 2  13213     Murray County, Georgia B28002_001    15246 277
## 3  13229     Pierce County, Georgia B28002_001     7605 289
## 4  13261     Sumter County, Georgia B28002_001    10986 420
## 5  13021       Bibb County, Georgia B28002_001    59921 748
## 6  13151      Henry County, Georgia B28002_001    86704 692
## 7  13195    Madison County, Georgia B28002_001    11242 275
## 8  13265 Taliaferro County, Georgia B28002_001      588 101
## 9  13267   Tattnall County, Georgia B28002_001     7630 278
## 10 13237     Putnam County, Georgia B28002_001    10435 356
##                          geometry
## 1  MULTIPOLYGON (((-83.10326 3...
## 2  MULTIPOLYGON (((-84.94433 3...
## 3  MULTIPOLYGON (((-82.41725 3...
## 4  MULTIPOLYGON (((-84.44381 3...
## 5  MULTIPOLYGON (((-83.89192 3...
## 6  MULTIPOLYGON (((-84.35419 3...
## 7  MULTIPOLYGON (((-83.40243 3...
## 8  MULTIPOLYGON (((-83.01087 3...
## 9  MULTIPOLYGON (((-82.23371 3...
## 10 MULTIPOLYGON (((-83.54582 3...
```

```
ga_internet_sf2017 <- get_acs(
  geography = "county",
  variables = "B28002_001E",
  state = "GA",
  year = 2017,
  geometry = TRUE
)
```

```
## Getting data from the 2013-2017 5-year ACS
```

```
## Downloading feature geometry from the Census website.  To cache shapefiles for use in future sessions
```

```
##   |                                                                              |
```

```
fulton_internet_sf <- get_acs(
  geography = "tract",
  variables = "B28002_013E",
  state = "GA",
  county = "Fulton",
  year = 2024,
  geometry = TRUE
)
```

```
## Getting data from the 2020-2024 5-year ACS
```

```
## Downloading feature geometry from the Census website.  To cache shapefiles for use in future sessions
```

```
##   |                                                                              |
```

```
fulton_internet_sf2017 <- get_acs(
  geography = "tract",
  variables = "B28002_013E",
  state = "GA",
  county = "Fulton",
  year = 2017,
  geometry = TRUE
)
```

```
## Getting data from the 2013-2017 5-year ACS
```

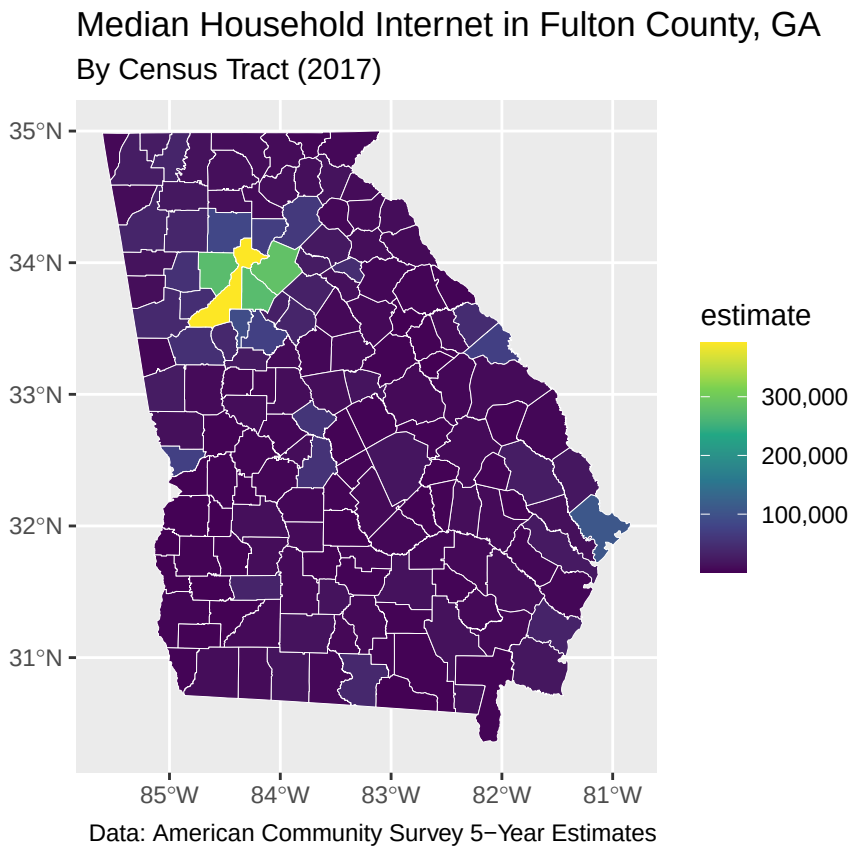
```
## Downloading feature geometry from the Census website.  To cache shapefiles for use in future sessions
```

```
##   |                                                                              |
```

```
ggplot(ga_internet_sf2017) +
  geom_sf(aes(fill = estimate), color = "white", size = 0.1) +
  scale_fill_viridis(option = "magma", name = "Median Household Internet") +
  labs(
    title = "Median Household Internet in Fulton County, GA",
    subtitle = "By Census Tract (2017)",
```

```
    caption = "Data: American Community Survey 5-Year Estimates"
  ) +
scale_fill_viridis_c(
  labels = scales::label_comma()
)
```

```
## Scale for fill is already present.
## Adding another scale for fill, which will replace the existing scale.
```




```
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12),
    axis.title = element_blank(),
    legend.position = "right"
  )
```

```
## <theme> List of 144
##  $ line                            : <ggplot2::element_line>
##   ..@ colour       : chr "black"
##   ..@ linewidth    : num 0.5
##   ..@ linetype     : num 1
##   ..@ lineend      : chr "butt"
##   ..@ linejoin     : chr "round"
```

```
##   ..@ arrow        : logi FALSE
##   ..@ arrow.fill   : chr "black"
##   ..@ inherit.blank: logi TRUE
##  $ rect                           : <ggplot2::element_rect>
##   ..@ fill         : chr "white"
##   ..@ colour       : chr "black"
##   ..@ linewidth    : num 0.5
##   ..@ linetype     : num 1
##   ..@ linejoin     : chr "round"
##   ..@ inherit.blank: logi TRUE
##  $ text                           : <ggplot2::element_text>
##   ..@ family       : chr ""
##   ..@ face         : chr "plain"
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : chr "black"
##   ..@ size         : num 11
##   ..@ hjust        : num 0.5
##   ..@ vjust        : num 0.5
##   ..@ angle        : num 0
##   ..@ lineheight   : num 0.9
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 0
##   ..@ debug        : logi FALSE
##   ..@ inherit.blank: logi TRUE
##  $ title                          : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ point                          : <ggplot2::element_point>
##   ..@ colour       : chr "black"
##   ..@ shape        : num 19
##   ..@ size         : num 1.5
##   ..@ fill         : chr "white"
##   ..@ stroke       : num 0.5
##   ..@ inherit.blank: logi TRUE
##  $ polygon                        : <ggplot2::element_polygon>
##   ..@ fill         : chr "white"
##   ..@ colour       : chr "black"
##   ..@ linewidth    : num 0.5
##   ..@ linetype     : num 1
##   ..@ linejoin     : chr "round"
##   ..@ inherit.blank: logi TRUE
```

```
##  $ geom                          : <ggplot2::element_geom>
##   ..@ ink        : chr "black"
##   ..@ paper      : chr "white"
##   ..@ accent     : chr "#3366FF"
##   ..@ linewidth  : num 0.5
##   ..@ borderwidth: num 0.5
##   ..@ linetype   : int 1
##   ..@ bordertype : int 1
##   ..@ family     : chr ""
##   ..@ fontsize   : num 3.87
##   ..@ pointsize  : num 1.5
##   ..@ pointshape : num 19
##   ..@ colour     : NULL
##   ..@ fill       : NULL
##  $ spacing                       : 'simpleUnit' num 5.5points
##   ..- attr(*, "unit")= int 8
##  $ margins                       : <ggplot2::margin> num [1:4] 5.5 5.5 5.5 5.5
##  $ aspect.ratio                  : NULL
##  $ axis.title                    : <ggplot2::element_blank>
##  $ axis.title.x                  : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 2.75 0 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.title.x.top              : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 0
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 2.75 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.title.x.bottom           : NULL
##  $ axis.title.y                  : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
```

```
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : num 90
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 2.75 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.title.y.left               : NULL
##  $ axis.title.y.right              : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : num -90
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 2.75
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text                       : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : chr "#4D4D4DFF"
##   ..@ size         : 'rel' num 0.8
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.x                     : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
```

```
##   ..@ margin       : <ggplot2::margin> num [1:4] 2.2 0 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.x.top                : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 4.95 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.x.bottom             : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 4.95 0 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.y                    : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : num 1
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 2.2 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.y.left               : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
```

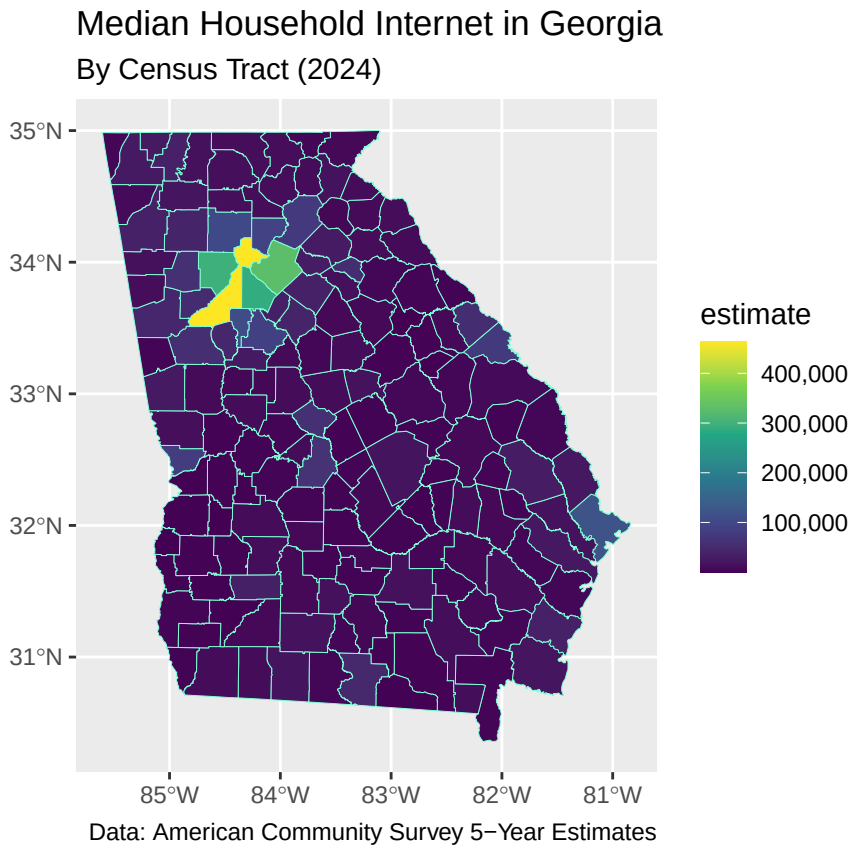
```
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 4.95 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.y.right               : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 4.95
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.theta                 : NULL
##  $ axis.text.r                     : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : num 0.5
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 2.2 0 2.2
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.ticks                      : <ggplot2::element_blank>
##  $ axis.ticks.x                    : NULL
##  $ axis.ticks.x.top                : NULL
##  $ axis.ticks.x.bottom             : NULL
##  $ axis.ticks.y                    : NULL
##  $ axis.ticks.y.left               : NULL
##  $ axis.ticks.y.right              : NULL
##  $ axis.ticks.theta                : NULL
##  $ axis.ticks.r                    : NULL
##  $ axis.minor.ticks.x.top          : NULL
##  $ axis.minor.ticks.x.bottom       : NULL
##  $ axis.minor.ticks.y.left         : NULL
##  $ axis.minor.ticks.y.right        : NULL
##  $ axis.minor.ticks.theta          : NULL
```

```
##  $ axis.minor.ticks.r             : NULL
##  $ axis.ticks.length              : 'rel' num 0.5
##  $ axis.ticks.length.x            : NULL
##  $ axis.ticks.length.x.top        : NULL
##  $ axis.ticks.length.x.bottom     : NULL
##  $ axis.ticks.length.y            : NULL
##  $ axis.ticks.length.y.left       : NULL
##  $ axis.ticks.length.y.right      : NULL
##  $ axis.ticks.length.theta        : NULL
##  $ axis.ticks.length.r            : NULL
##  $ axis.minor.ticks.length        : 'rel' num 0.75
##  $ axis.minor.ticks.length.x      : NULL
##  $ axis.minor.ticks.length.x.top  : NULL
##  $ axis.minor.ticks.length.x.bottom: NULL
##  $ axis.minor.ticks.length.y      : NULL
##  $ axis.minor.ticks.length.y.left : NULL
##  $ axis.minor.ticks.length.y.right : NULL
##  $ axis.minor.ticks.length.theta  : NULL
##  $ axis.minor.ticks.length.r      : NULL
##  $ axis.line                      : <ggplot2::element_blank>
##  $ axis.line.x                    : NULL
##  $ axis.line.x.top                : NULL
##  $ axis.line.x.bottom             : NULL
##  $ axis.line.y                    : NULL
##  $ axis.line.y.left               : NULL
##  $ axis.line.y.right              : NULL
##  $ axis.line.theta                : NULL
##  $ axis.line.r                    : NULL
##  $ legend.background              : <ggplot2::element_blank>
##  $ legend.margin                  : NULL
##  $ legend.spacing                 : 'rel' num 2
##  $ legend.spacing.x               : NULL
##  $ legend.spacing.y               : NULL
##  $ legend.key                     : <ggplot2::element_blank>
##  $ legend.key.size                : 'simpleUnit' num 1.2lines
##   ..- attr(*, "unit")= int 3
##  $ legend.key.height              : NULL
##  $ legend.key.width               : NULL
##  $ legend.key.spacing             : NULL
##  $ legend.key.spacing.x           : NULL
##  $ legend.key.spacing.y           : NULL
##  $ legend.key.justification       : NULL
##  $ legend.frame                   : NULL
##  $ legend.ticks                   : NULL
##  $ legend.ticks.length            : 'rel' num 0.2
##  $ legend.axis.line               : NULL
##  $ legend.text                    : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : 'rel' num 0.8
```

```
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ legend.text.position            : NULL
##  $ legend.title                    : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : num 0
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ legend.title.position           : NULL
##  $ legend.position                 : chr "right"
##  $ legend.position.inside          : NULL
##  $ legend.direction                : NULL
##  $ legend.byrow                    : NULL
##  $ legend.justification            : chr "center"
##  $ legend.justification.top        : NULL
##  $ legend.justification.bottom     : NULL
##  $ legend.justification.left       : NULL
##  $ legend.justification.right      : NULL
##  $ legend.justification.inside     : NULL
##   [list output truncated]
##  @ complete: logi TRUE
##  @ validate: logi TRUE
```

```
ggplot(ga_internet_sf) +
  geom_sf(aes(fill = estimate), color = "aquamarine", size = 0.1) +
  scale_fill_viridis(option = "magma", name = "Median Household Internet") +
  labs(
    title = "Median Household Internet in Georgia",
    subtitle = "By Census Tract (2024)",
    caption = "Data: American Community Survey 5-Year Estimates"
  ) +
scale_fill_viridis_c(
  labels = scales::label_comma()
)
```

```
## Scale for fill is already present.
## Adding another scale for fill, which will replace the existing scale.
```




```
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12),
    axis.title = element_blank(),
    legend.position = "right"
  )
```

```
## <theme> List of 144
##  $ line                            : <ggplot2::element_line>
##   ..@ colour       : chr "black"
##   ..@ linewidth    : num 0.5
##   ..@ linetype     : num 1
##   ..@ lineend      : chr "butt"
##   ..@ linejoin     : chr "round"
##   ..@ arrow        : logi FALSE
##   ..@ arrow.fill   : chr "black"
##   ..@ inherit.blank: logi TRUE
##  $ rect                            : <ggplot2::element_rect>
##   ..@ fill         : chr "white"
##   ..@ colour       : chr "black"
##   ..@ linewidth    : num 0.5
##   ..@ linetype     : num 1
##   ..@ linejoin     : chr "round"
##   ..@ inherit.blank: logi TRUE
##  $ text                            : <ggplot2::element_text>
```

```
##   ..@ family       : chr ""
##   ..@ face         : chr "plain"
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : chr "black"
##   ..@ size         : num 11
##   ..@ hjust        : num 0.5
##   ..@ vjust        : num 0.5
##   ..@ angle        : num 0
##   ..@ lineheight   : num 0.9
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 0
##   ..@ debug        : logi FALSE
##   ..@ inherit.blank: logi TRUE
##  $ title                          : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ point                          : <ggplot2::element_point>
##   ..@ colour       : chr "black"
##   ..@ shape        : num 19
##   ..@ size         : num 1.5
##   ..@ fill         : chr "white"
##   ..@ stroke       : num 0.5
##   ..@ inherit.blank: logi TRUE
##  $ polygon                        : <ggplot2::element_polygon>
##   ..@ fill         : chr "white"
##   ..@ colour       : chr "black"
##   ..@ linewidth    : num 0.5
##   ..@ linetype     : num 1
##   ..@ linejoin     : chr "round"
##   ..@ inherit.blank: logi TRUE
##  $ geom                           : <ggplot2::element_geom>
##   ..@ ink        : chr "black"
##   ..@ paper      : chr "white"
##   ..@ accent     : chr "#3366FF"
##   ..@ linewidth  : num 0.5
##   ..@ borderwidth: num 0.5
##   ..@ linetype   : int 1
##   ..@ bordertype : int 1
##   ..@ family     : chr ""
##   ..@ fontsize   : num 3.87
##   ..@ pointsize  : num 1.5
```

```
##   ..@ pointshape : num 19
##   ..@ colour     : NULL
##   ..@ fill       : NULL
##  $ spacing                       : 'simpleUnit' num 5.5points
##   ..- attr(*, "unit")= int 8
##  $ margins                       : <ggplot2::margin> num [1:4] 5.5 5.5 5.5 5.5
##  $ aspect.ratio                  : NULL
##  $ axis.title                    : <ggplot2::element_blank>
##  $ axis.title.x                  : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 2.75 0 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.title.x.top              : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 0
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 2.75 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.title.x.bottom           : NULL
##  $ axis.title.y                  : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : num 90
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 2.75 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
```

```
##  $ axis.title.y.left               : NULL
##  $ axis.title.y.right              : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : num -90
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 2.75
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text                       : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : chr "#4D4D4DFF"
##   ..@ size         : 'rel' num 0.8
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.x                     : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 2.2 0 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.x.top                 : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
```

```
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 4.95 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.x.bottom             : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 4.95 0 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.y                    : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : num 1
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 2.2 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.y.left               : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 4.95 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.y.right              : <ggplot2::element_text>
##   ..@ family       : NULL
```

```
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 4.95
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.theta                : NULL
##  $ axis.text.r                    : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : num 0.5
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 2.2 0 2.2
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.ticks                     : <ggplot2::element_blank>
##  $ axis.ticks.x                   : NULL
##  $ axis.ticks.x.top               : NULL
##  $ axis.ticks.x.bottom            : NULL
##  $ axis.ticks.y                   : NULL
##  $ axis.ticks.y.left              : NULL
##  $ axis.ticks.y.right             : NULL
##  $ axis.ticks.theta               : NULL
##  $ axis.ticks.r                   : NULL
##  $ axis.minor.ticks.x.top         : NULL
##  $ axis.minor.ticks.x.bottom      : NULL
##  $ axis.minor.ticks.y.left        : NULL
##  $ axis.minor.ticks.y.right       : NULL
##  $ axis.minor.ticks.theta         : NULL
##  $ axis.minor.ticks.r             : NULL
##  $ axis.ticks.length              : 'rel' num 0.5
##  $ axis.ticks.length.x            : NULL
##  $ axis.ticks.length.x.top        : NULL
##  $ axis.ticks.length.x.bottom     : NULL
##  $ axis.ticks.length.y            : NULL
##  $ axis.ticks.length.y.left       : NULL
##  $ axis.ticks.length.y.right      : NULL
##  $ axis.ticks.length.theta        : NULL
##  $ axis.ticks.length.r            : NULL
##  $ axis.minor.ticks.length        : 'rel' num 0.75
```

```
##  $ axis.minor.ticks.length.x       : NULL
##  $ axis.minor.ticks.length.x.top   : NULL
##  $ axis.minor.ticks.length.x.bottom: NULL
##  $ axis.minor.ticks.length.y       : NULL
##  $ axis.minor.ticks.length.y.left  : NULL
##  $ axis.minor.ticks.length.y.right : NULL
##  $ axis.minor.ticks.length.theta   : NULL
##  $ axis.minor.ticks.length.r       : NULL
##  $ axis.line                       : <ggplot2::element_blank>
##  $ axis.line.x                     : NULL
##  $ axis.line.x.top                 : NULL
##  $ axis.line.x.bottom              : NULL
##  $ axis.line.y                     : NULL
##  $ axis.line.y.left                : NULL
##  $ axis.line.y.right               : NULL
##  $ axis.line.theta                 : NULL
##  $ axis.line.r                     : NULL
##  $ legend.background               : <ggplot2::element_blank>
##  $ legend.margin                   : NULL
##  $ legend.spacing                  : 'rel' num 2
##  $ legend.spacing.x                : NULL
##  $ legend.spacing.y                : NULL
##  $ legend.key                      : <ggplot2::element_blank>
##  $ legend.key.size                 : 'simpleUnit' num 1.2lines
##   ..- attr(*, "unit")= int 3
##  $ legend.key.height               : NULL
##  $ legend.key.width                : NULL
##  $ legend.key.spacing              : NULL
##  $ legend.key.spacing.x            : NULL
##  $ legend.key.spacing.y            : NULL
##  $ legend.key.justification        : NULL
##  $ legend.frame                    : NULL
##  $ legend.ticks                    : NULL
##  $ legend.ticks.length             : 'rel' num 0.2
##  $ legend.axis.line                : NULL
##  $ legend.text                     : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : 'rel' num 0.8
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ legend.text.position            : NULL
##  $ legend.title                    : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
```

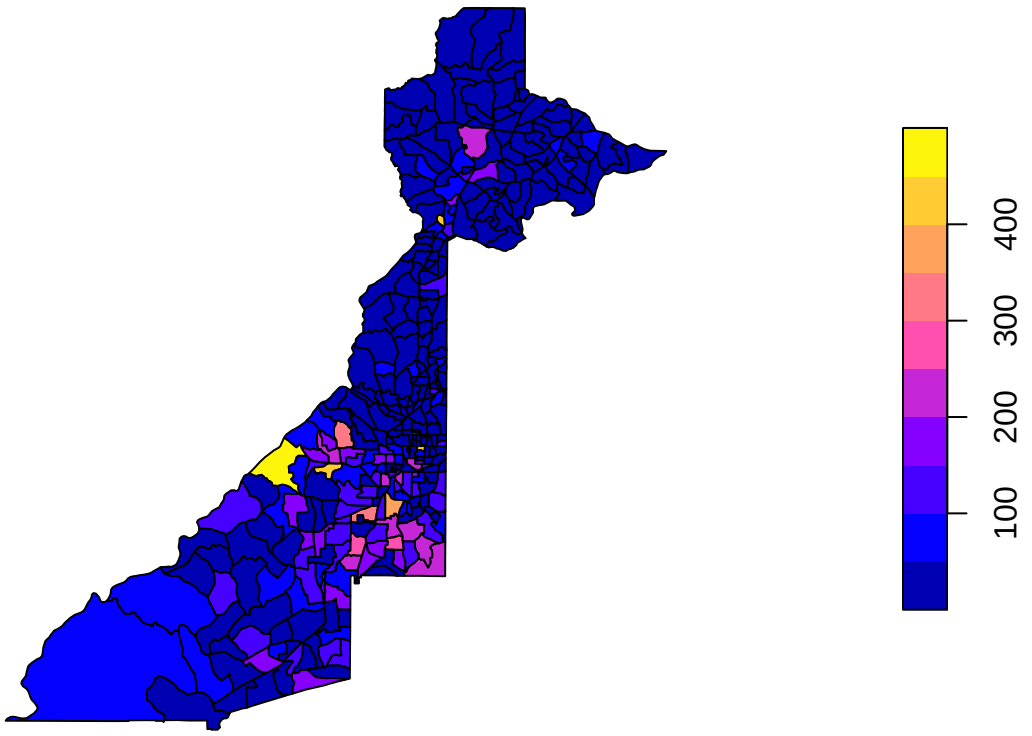
```
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : num 0
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ legend.title.position           : NULL
##  $ legend.position                 : chr "right"
##  $ legend.position.inside          : NULL
##  $ legend.direction                : NULL
##  $ legend.byrow                    : NULL
##  $ legend.justification            : chr "center"
##  $ legend.justification.top        : NULL
##  $ legend.justification.bottom     : NULL
##  $ legend.justification.left       : NULL
##  $ legend.justification.right      : NULL
##  $ legend.justification.inside     : NULL
##   [list output truncated]
##  @ complete: logi TRUE
##  @ validate: logi TRUE
```

```
fulton_blackpop <- get_acs(
  geography = "tract",
  variables = "B01001B_001", # Black population variable
  state = "GA",
  county = "Fulton",
  year = 2024,
  geometry = T
)
```

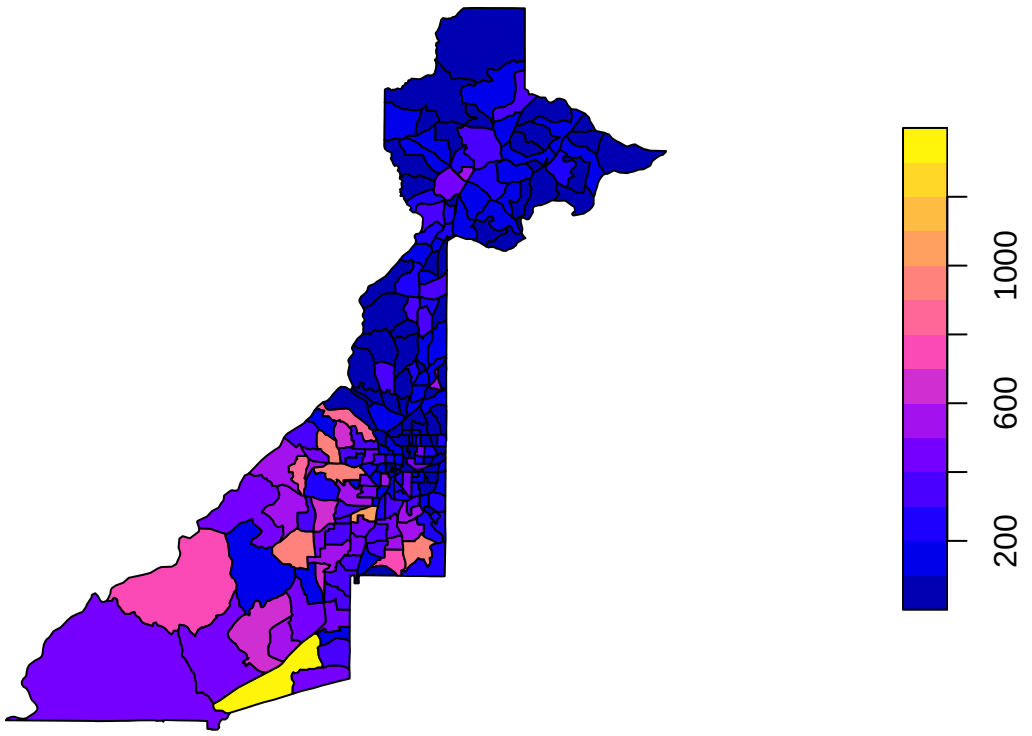
```
## Getting data from the 2020-2024 5-year ACS
```

```
## Downloading feature geometry from the Census website.  To cache shapefiles for use in future sessions
```

```
plot(fulton_internet_sf['estimate'])
```



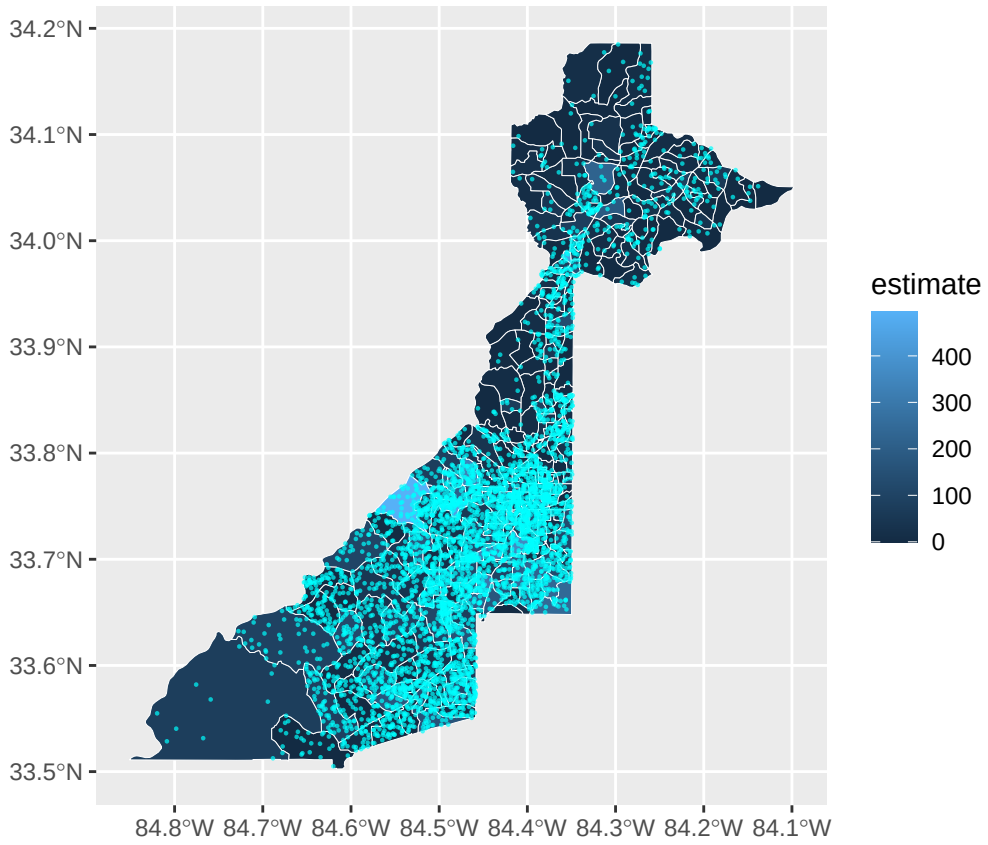

```
plot(fulton_internet_sf2017['estimate'])
```



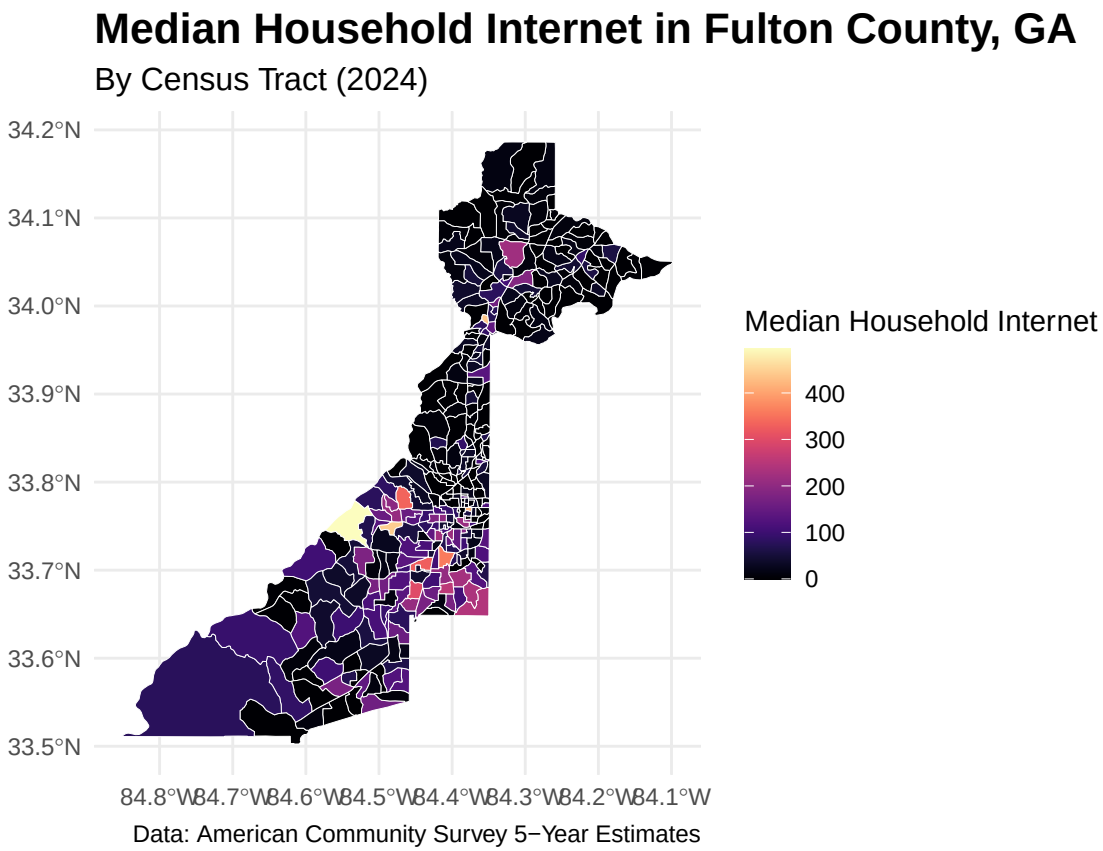

```
blackpop_dots <- as_dot_density(
  fulton_blackpop,
  value = "estimate",
  values_per_dot = 100 # adjust this number to
)
```

```
ggplot() +
# Base layer: internet
geom_sf(
data = fulton_internet_sf,
aes(fill = estimate),
color = "white",
linewidth = 0.1
) +

# Overlay: dot density
geom_sf(
data = blackpop_dots,
color = "cyan",
size = 0.15,
alpha = 0.7,
inherit.aes = FALSE
)
```




```
ggplot(fulton_internet_sf) +
  geom_sf(aes(fill = estimate), color = "white", size = 0.1) +
  scale_fill_viridis(option = "magma", name = "Median Household Internet") +
  labs(
    title = "Median Household Internet in Fulton County, GA",
    subtitle = "By Census Tract (2024)",
    caption = "Data: American Community Survey 5-Year Estimates"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12),
    axis.title = element_blank(),
    legend.position = "right"
  )
```


Median Household Internet in Fulton County, GA
By Census Tract (2024)
34.2°N
34.1°N
34.0°N
33.9°N
33.8°N
33.7°N
33.6°N
33.5°N
84.8°W
84.7°W
84.6°W
84.5°W
84.4°W
84.3°W
84.2°W
84.1°W
Median Household Internet
400
300
200
100
0
Data: American Community Survey 5−Year Estimates